

Over 400-mW O-band Monolithic MOPA Laser Array Based on REC Technology for CPO/OIO

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Abstract We propose and demonstrate a MOPA array with precise wavelength spacing based on the REC technique. The MOPA laser array achieves an output power exceeding 400 mW when operating at 50 °C, providing an excellent high-power light source solution for CPO/ OIO technologies. ©2026 The Author(s)

Introduction

Recently, highly integrated optical interconnects, such as co-packaged optics (CPO) and optical I/O (OIO), have been extensively investigated for high-performance computing [1], [2], [3]. In these systems, large insertion losses caused by silicon photonic components strictly limit the optical link budget. Therefore, multi-wavelength external light sources with a continuous-wave (CW) output power of over 400 mW/channel are required to increase the aggregate data throughput [4], [5], [6].

In this paper, we propose a monolithic master oscillator power amplifier (MOPA) array to meet this requirement. The master oscillator (MO) integrates a reconstruction-equivalent-chirp (REC)-based equivalent asymmetric corrugated pitch modulation (CPM) grating. This grating structure flattens the longitudinal photon density distribution, thereby suppressing the spatial hole burning effect. Furthermore, an asymmetric waveguide is adopted to shift the optical mode toward the n-type substrate, which reduces the free-carrier absorption loss. These structural designs, combined with a buried heterostructure (BH) and an aluminum-free active region, provide high injection efficiency and long-term reliability.

We fabricated a 12-channel O-band MOPA laser array with a 400-GHz channel spacing. At an ambient temperature of 50 °C, a CW output power exceeding 400 mW/channel and a side-mode suppression ratio (SMSR) of over 50 dB were obtained. An intrinsic linewidth of 152.5 kHz was measured at a total injection current of 1000 mA. In addition, a nearly symmetric far-field divergence ($16.8^\circ \times 18.9^\circ$) was confirmed. These results demonstrate the suitability of the proposed MOPA array as an efficient light source for advanced CPO and OIO applications.

Device design and fabrication

The master oscillator (MO) employs a REC-based equivalent asymmetric CPM grating [7]. This structure flattens the intracavity photon density to suppress spatial hole burning, achieving mode-hop-free operation. It also compensates for the random phase at the HR facet to improve the single-mode yield [7]. Furthermore, the REC technique enables grating fabrication via standard holographic exposure and photolithography without using EBL. This approach provides precise channel spacing and improves the fabrication tolerance by two orders of magnitude compared to EBL [8], [9], [10]. Figures 1(a) and 1(b) show the schematic and simulated transmission spectrum of the grating, respectively.

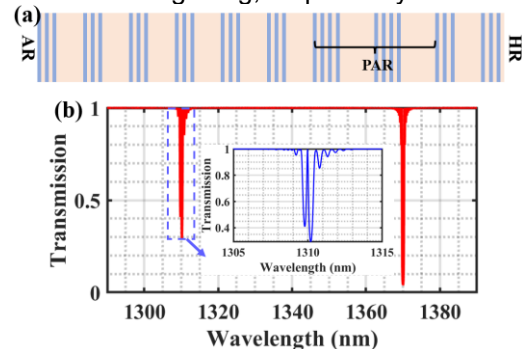


Fig. 1: (a) Schematic of the sampled grating based on REC technique, and (b) simulated transmission spectra of the grating using the REC technique.

Figure 2(a) shows a cross-sectional SEM image of the fabricated BH waveguide. The active region consists of four 1.0% compressive-strained InGaAsP wells and five 0.3% tensile-strained barriers. An n-InGaAsP layer is incorporated to shift the optical mode toward the n-region, reducing p-side free-carrier absorption and vertical divergence for efficient coupling. An aluminum-free material system is employed to ensure

long-term reliability. Fig. 1(b) shows an SEM image of the REC-based sampled grating.

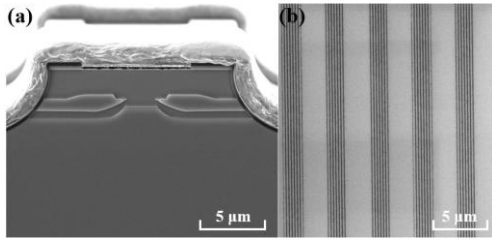


Fig. 2: (a) SEM image of the cross-sectional view of the fabricated BH waveguide and (b) SEM image of the sampled grating structure.

Figure 3 shows an optical microscope image of the fabricated array. The 12-channel chip has a footprint of 3 mm×2 mm with a channel pitch of 250 μm. Each MOPA element has a length of 2 mm, consisting of a 1-mm MO section and a 1-mm PA section. High-reflection (HR) and anti-reflection (AR) coatings were applied to the rear and front facets, respectively, to increase the output power. Additionally, a 7° tilted waveguide was formed at the PA facet to suppress optical reflections.

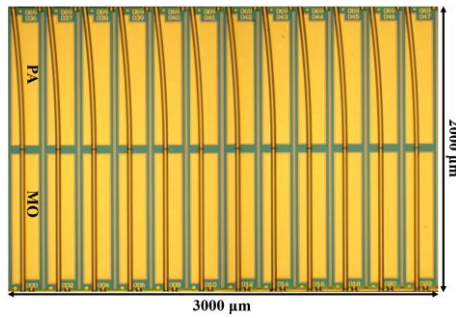


Fig. 3: Microscope image of the fabricated 12-channel MOPA laser array chip.

Device characterization

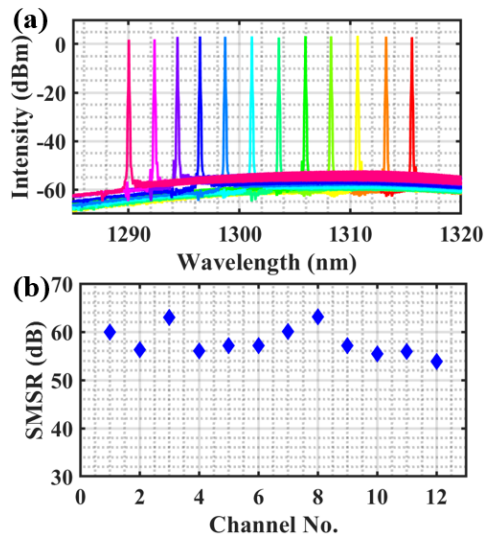


Fig. 4: Superimposed spectra of the 12-channel array at 50°C (MO and PA biased at 300 mA). (b) SMSR distribution demonstrating >50 dB across all channels.

Figure 4(a) shows the superimposed optical spectra of the 12-channel array measured at 50 °C. At an injection current of 300 mA for both the MO and PA sections, uniform channel spacing and single-mode operation were observed. Figure 3(b) shows the SMSR distribution of the 12 channels. An SMSR of over 50 dB was achieved for all channels.

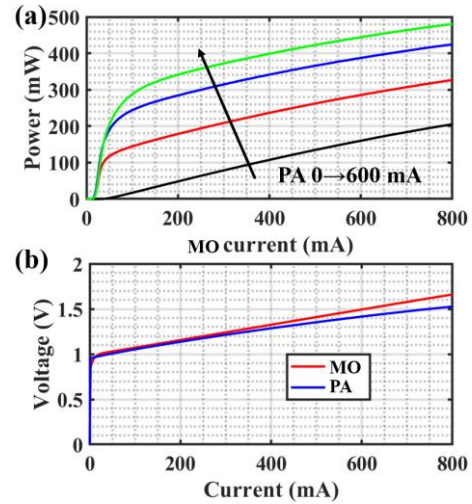


Fig. 5: Measured (a) P-I curves under various PA currents and (b) I-V characteristics of the device at 50°C.

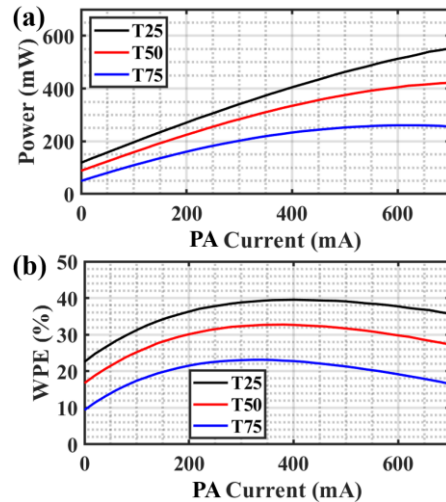


Fig. 6: Measured (a) output power and (b) WPE versus PA current at varying temperatures (25°C to 75°C), with a fixed MO bias of 400 mA.

The PIV characteristics of the fabricated device were evaluated at a stabilized temperature of 50 °C. Figure 5(a) shows the continuous-wave (CW) P-I curves measured under various PA bias currents from 0 to 600 mA. The threshold current of the MO section is observed to be less than 50 mA. The optical output power increases significantly with the PA bias current, reaching 400 mW at an PA current of 600 mA and an MO current of 400 mA. The measured I-V characteristics are displayed in Fig. 5(b), indicating low differential resistances of 710 mΩ and 860 mΩ for the PA

and MO sections, respectively.

The temperature dependence of the MOPA laser array was also evaluated. With the injection current to the MO fixed at 400 mA, the output power and wall-plug efficiency (WPE) as a function of the PA current were measured. Figures 6(a) and 6(b) show the results at 25, 50, and 75 °C. At 25 °C, the maximum output power exceeded 550 mW, with a peak WPE of approximately 40%. Even at a high temperature of 75 °C, an output power of over 250 mW and a peak WPE of over 20% were successfully maintained.

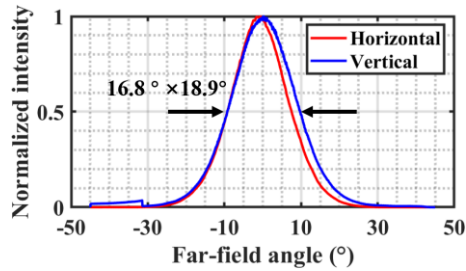


Fig. 7. Far-field divergence curves (300 mA, 50 °C) exhibiting FWHMs of 16.8° (horizontal) and 18.9° (vertical).

Figure 7 shows the far-field divergence curves measured at 300 mA and 50 °C. The full width at half maximum (FWHM) angles in the horizontal and vertical directions were evaluated to be 16.8° and 18.9°, respectively. This narrow and nearly symmetric beam profile is advantageous for achieving high-efficiency optical coupling.

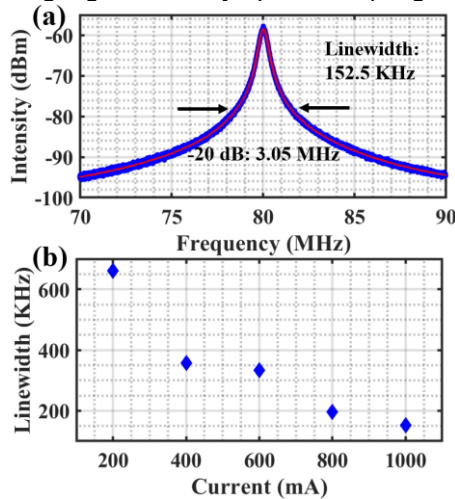


Fig. 8: (a) RF beat spectrum with a total injection current of 1000 mA. (b) Intrinsic linewidth as a function of the total injection current ranging from 200 to 1000 mA.

The spectral linewidth was evaluated using the delayed self-heterodyne (DSH) method at a constant temperature of 25 °C. In this measurement, the MO and PA sections were simultaneously driven by a single-channel low-noise current source. Figure 8(a) shows the measured beat spectrum at a total injection current of 1000 mA. The 20-dB bandwidth was observed to be 3.05 MHz, which corresponds to an intrinsic

Lorentzian linewidth of 152.5 kHz. Figure 8(b) shows the dependence of the linewidth on the total injection current. As the total current increased from 200 to 1000 mA, the linewidth decreased from 660.8 to 152.5 kHz.

Conclusions

We demonstrated a 12-channel O-band monolithic MOPA laser array for CPO/OIO technologies. At 50 °C, the array exhibited a CW output power of >400 mW/channel, an SMSR of >50 dB, and a uniform 400-GHz spacing. Additionally, a narrow intrinsic linewidth of 152.5 kHz and a nearly symmetric far-field profile of 16.8°×18.9° were confirmed. These results indicate that this cost-effective architecture is highly suitable as a light source for future integrated photonics.

Acknowledgements

National Key Research and Development Program of China (2023YFB2806400); National Natural Science Foundation of China (62404098); Natural Science Foundation of Jiangsu Province (BK20241196).

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